Tutorial: Instrumental Variable Method

1 Overview

This tutorial introduces **instrumental variable methods** by replicating some of the key results in the paper:

Acemoglu, Daron, Simon Johnson, and James A. Robinson. 2001. "The Colonial Origins of Comparative Development: An Empirical Investigation." American Economic Review 91 (5): 1369–1401. [link] [AJR]

1.1 Learning Objectives

By the end of this tutorial, you should:

- have a good understanding of the main assumptions underlying instrumental variable method
- be able to estimate IV model from scratch
- compare IV estimates with OLS
- conduct robustness checks to ensure your model is valid

1.2 Pre-requisites

You must attempt this exercise prior to your tutorial. To make the most out of this session, you should:

- Download the relevant files (data and R script)
- Read the introduction section of the paper
- Run the R script and understand each step

2 Task

1. Read the introduction of the AJR paper and answer the following questions:
   (a) Briefly discuss the main insight from the paper.
   (b) Estimate the following equation using OLS:

$$\log y_i = \mu + \alpha R_i + \mathbf{X}_i' \gamma + u_i$$

where,

- y_i is income per capita (`logpgp95`)
- R_i protection against expropriation measure (`avexpr`)

- $\mathbf{X}_i$ vector of other variables (`lat_abst`, `f_brit`, `f_french`, `sjlofr`, `catho80`, `muslim80` and `no_cpm80`)
- u_i is random error.

(c) Can we rely on the ols estimate $\hat{\alpha}$ in (b)? Briefly explain.
(d) Estimate the above equation by TSLS using log settler mortality (`logem4`) as an instrument for R_i (`avexpr`).
(e) Test for relevance of `logem4` as an instrument for `avexpr` using F-test and the robust Anderson-Rubin test.
(f) Does `logem4` satisfy instrument exogeneity? Discuss.